

Collections and Generics in C#

Engineering Solutions Using
Associative Collections



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In This Module...

Demo 1

Transparent
object cache

Demo 2

Domain-oriented
class

Utility class

Utility class

Utility class

Common
collection

Common
collection



In This Module...

Demo 1

Transparent
object cache

Caching solution 1

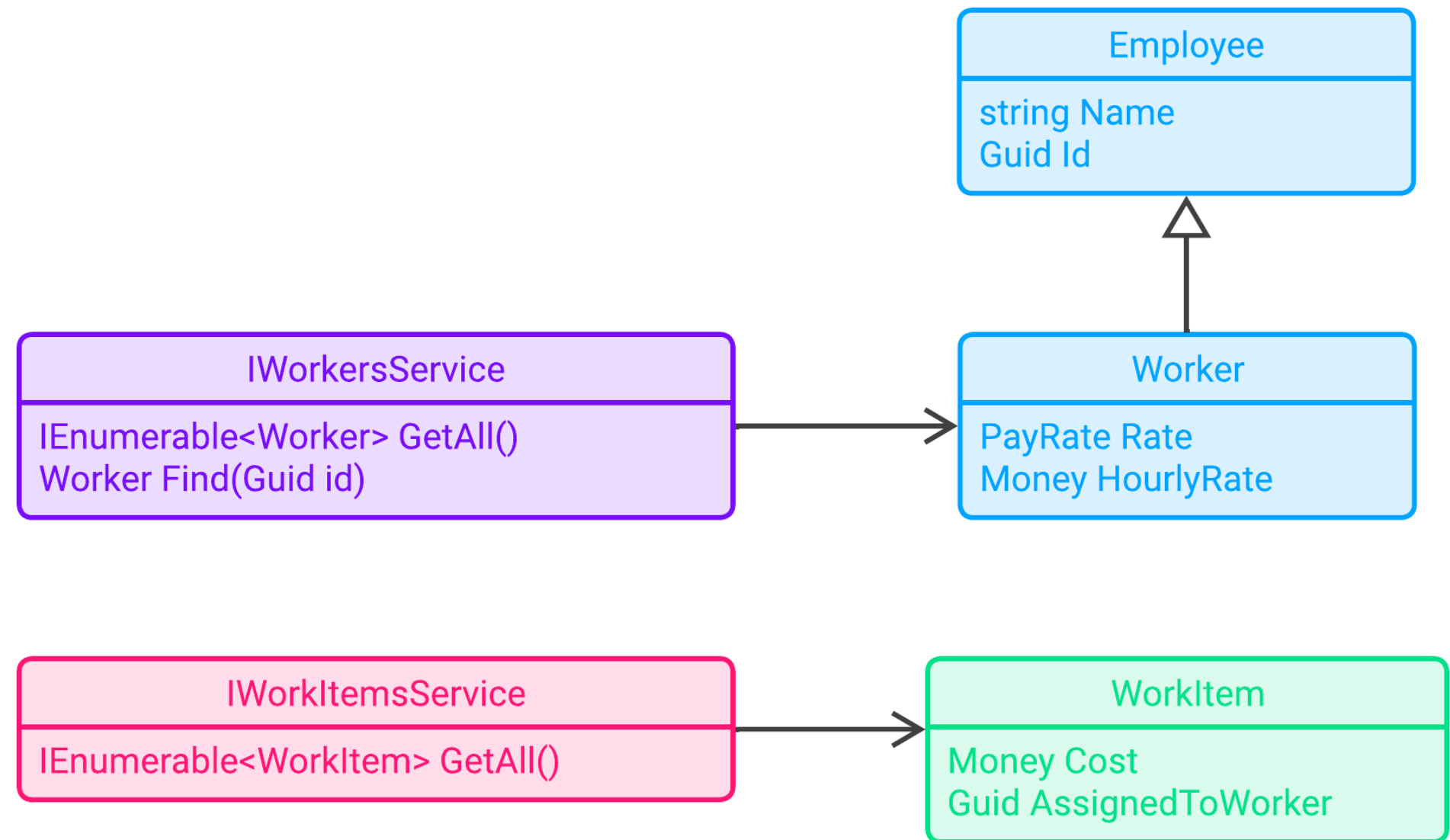
- Use a `HashSet<T>`
- Test if an object is a duplicate

Caching solution 2

- Use a `Dictionary<TKey, TValue>`
- Use constructor arguments as the key



Demo

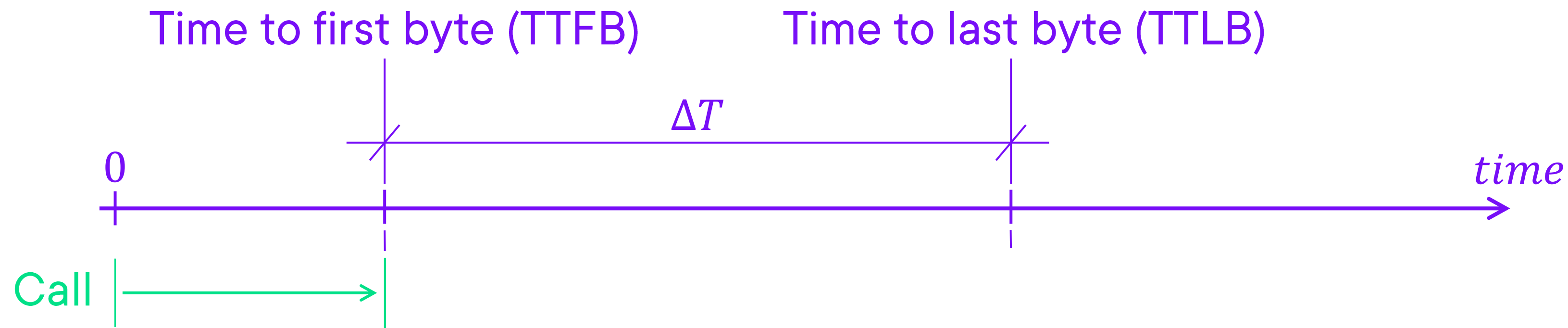


Use the IWorkItemsService to fetch work items

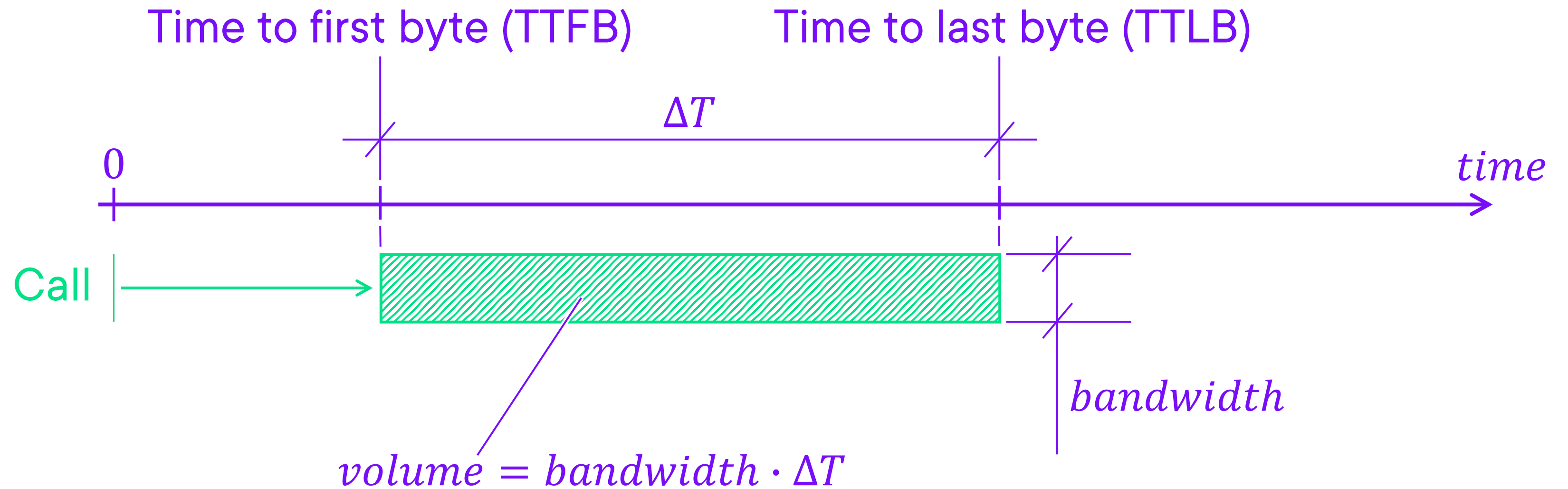
Use the IWorkersService to fetch the worker for each work item



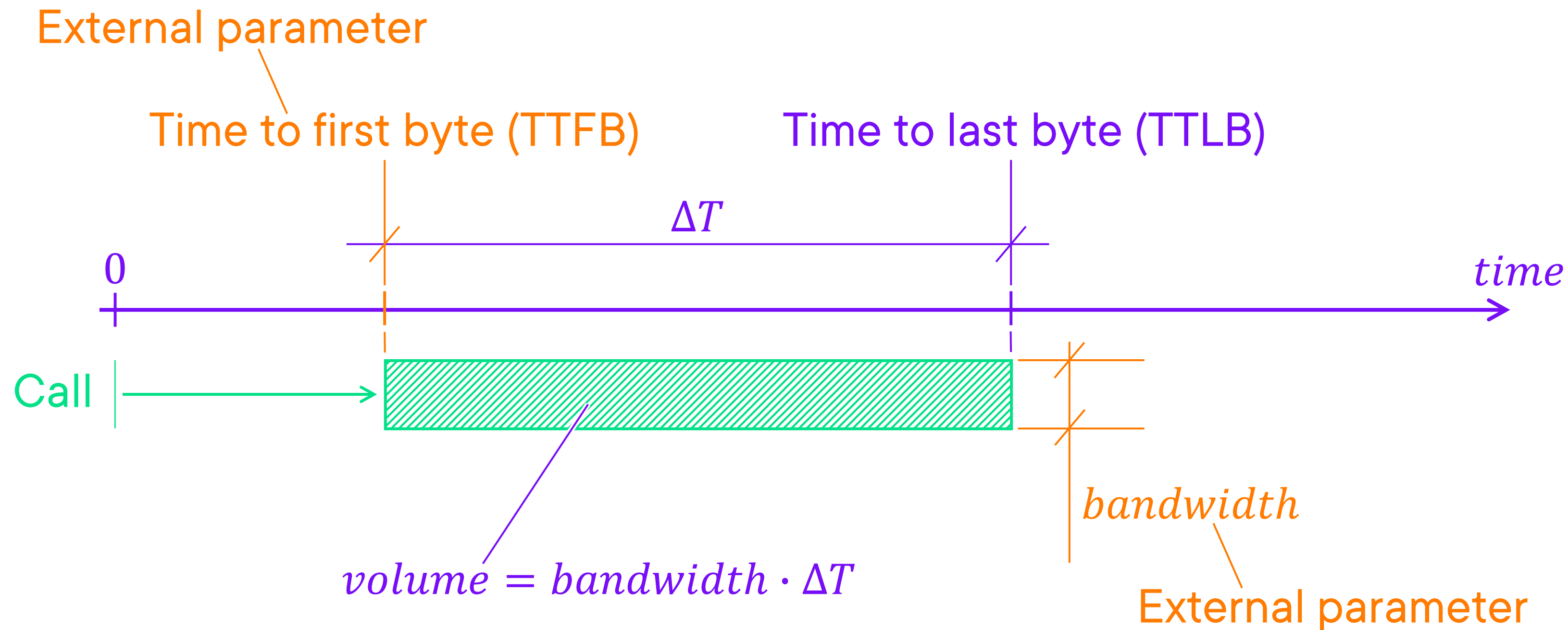
Service Communication Metrics



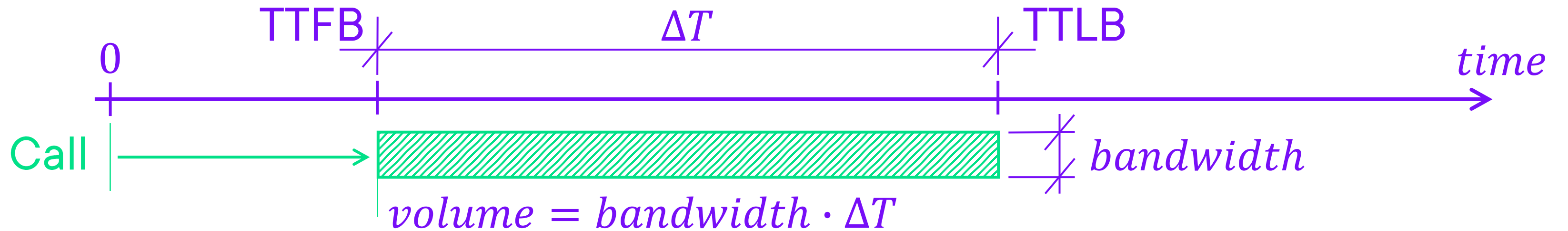
Service Communication Metrics



Service Communication Metrics



Service Communication Metrics



Chunky communication

Large volume implies $\Delta T \gg TTFB$

$$effective\ bandwidth \approx \frac{volume}{\Delta T} = bandwidth$$

Chatty communication

Small volume implies $TTFB \gg \Delta T$

$$effective\ bandwidth \approx \frac{volume}{TTFB} \ll bandwidth$$



- > .vscode
- ▼ ConsoleDemo
 - ConsoleDemo.csproj
 - C# CurrencyCreationBen...
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- ▼ Models
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 - C# FullySortedList.cs
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 - C# Currency.cs
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 - C# Money.cs
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 - C# PayRate.cs
 - C# Worker.cs
 - C# WorkItem.cs
 - ▼ Models.Tests
 - > Data
 - ▼ Services
 - C# DelayingWorkersSer...
 - C# TestWorkItemsServic...
 - Models.Tests.csproj

ConsoleDemo > C# Program.cs

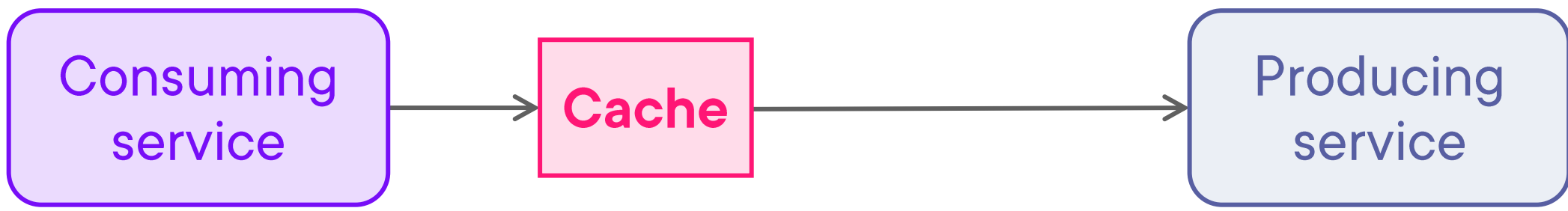
```
1 using System.Diagnostics;
2
3 try
4 {
5     (IWorkersService workers, IWorkItemsService workItems) = CreateServices();
6
7     int batchSize = 1000;
8
9     Stopwatch time = Stopwatch.StartNew();
10    foreach (WorkItem workItem in workItems.GetAll().Take(batchSize))
11    {
12        workers.Find(workItem.AssignedToWorker);
13    }
14    Console.WriteLine($"Processed {batchSize} item(s) in {time.Elapsed}");
15 }
```



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ConsoleDemo > Program.cs

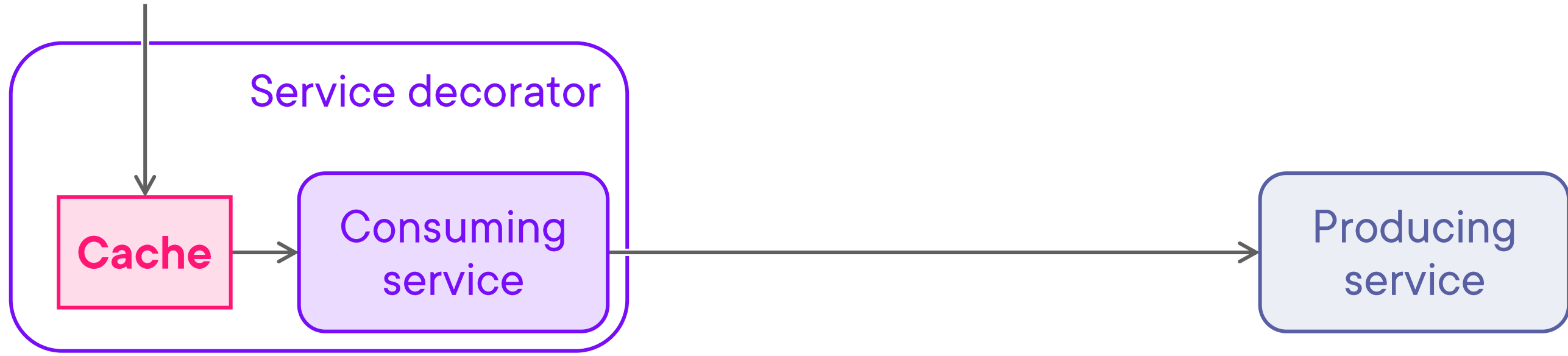
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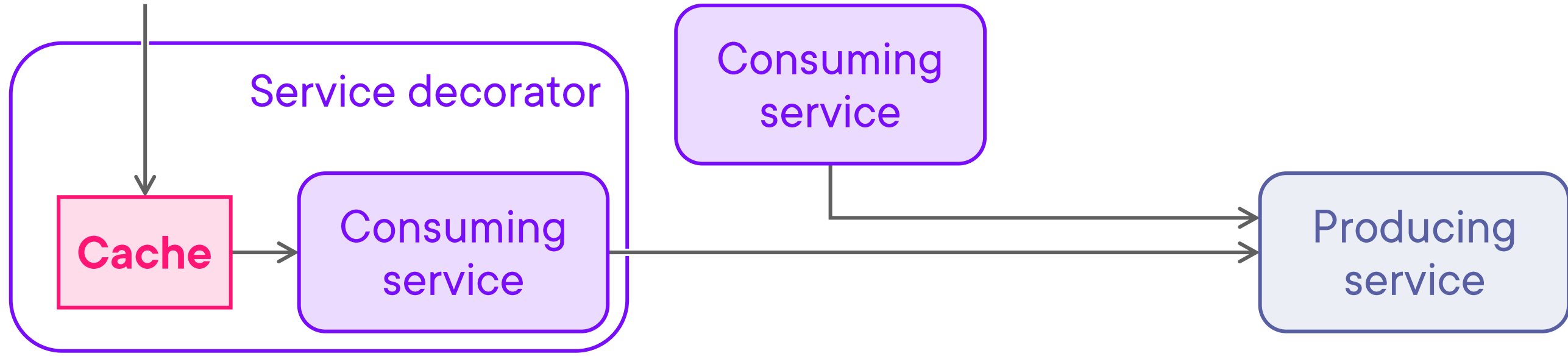
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Models > Collections > LruCache.cs > {} Models.Collections > Models.Collections.LruCache<TKey, TValue>

```
1 namespace Models.Collections;
2
3 public class LruCache<TKey, TValue> where TKey : IEquatable<TKey>
4 {
5 }
```

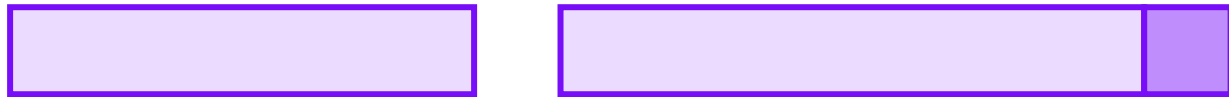
Access an item



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- ConsoleDemo
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 - Program.cs
- Models
 - Collections
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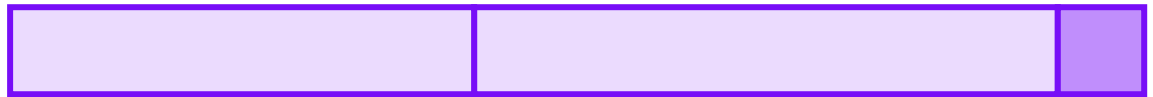
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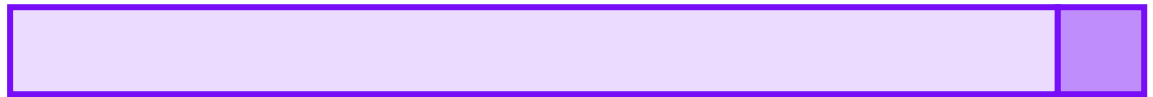
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 - Program.cs
- Models
 - Collections
 - FullySortedList.cs
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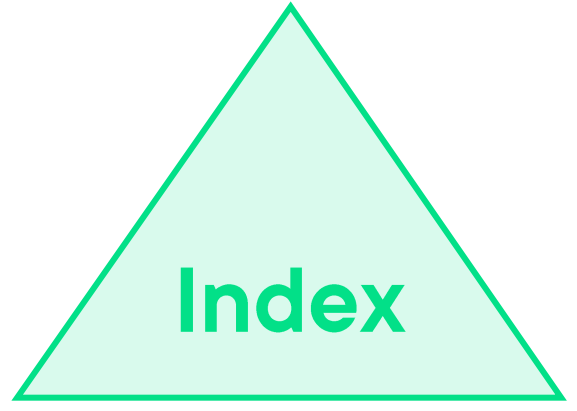


The least recently used item

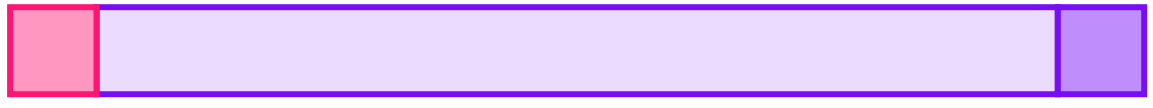
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↓ Maps key to location

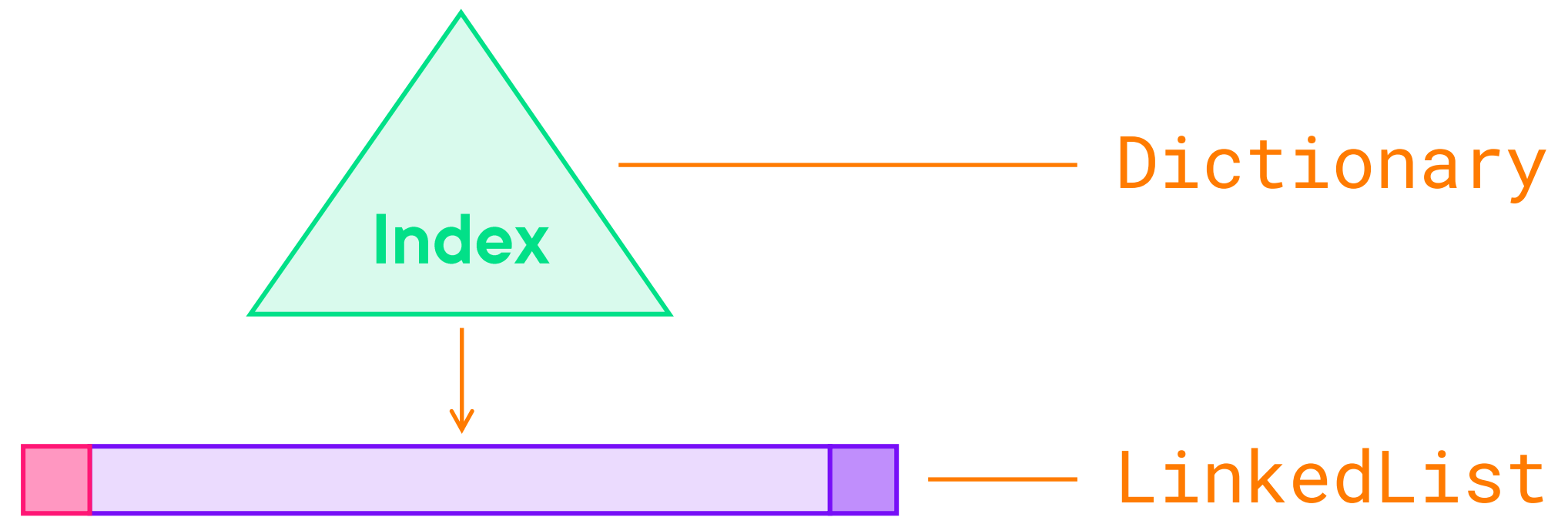


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Summary



Applied OOD to associative collections

Used a HashSet to share equatable objects

- Caused lower memory footprint
- Caused shorter execution time
- Designed a cache based on a HashSet

Designed a general-purpose LRU cache

- Cache keeps record of costly objects
- Used LRU cache as a private component



Summary



Caching is a matter of configuration

- Used object composition to inject a cache
- Consumer never depends on a collection

A digression on communication protocols

- Identified chatty as a problem
- Applied a local cache



Summary



Is Dictionary only a cache?

- Promises $O(1)$ time for operations
- Outperforms any linear collection
- Its unique ability is to *seek* by the key
- Try to view dictionary as a temporary cache



Up Next:

Engineering Custom Linear Collections

